

MATH3550 Real Analysis II

Midterm Mock

October 2026

There are three questions in this paper. You must complete all three questions to achieve full marks. The mock is worth 30 marks.

Duration: 90 minutes

1. (a) State the *classification of discontinuities of the first and second kind*. Further classify discontinuities of the first kind. (3)

- (b) Classify the discontinuities of the function $f(x) = \cos\left(\frac{\pi}{x}\right)$ at all points in $\mathbb{R}$, including the points at which f is not defined. (3)

- (c) Find the discontinuities of the function (4)

$$f(x) = \begin{cases} \frac{x^3}{x} & \text{if } x \leq 0, \\ x - \lfloor x \rfloor & \text{if } x > 0, \end{cases}$$

at all points in $\mathbb{R}$. Here $\lfloor x \rfloor$ is the floor function, returning the largest integer less than or equal to x . If the discontinuities are removable, remove them.

2. (a) State the definition of *uniform continuity*. (3)

- (b) Prove, using the ε - δ definition, that $\sin\left(\frac{1}{x}\right)$ is continuous on $(0, 1)$. (4)

- (c) Prove that $\sin\left(\frac{1}{x}\right)$ is not uniformly continuous on $(0, 1)$. (3)

3. (a) Let g be a function differentiable on an open interval $A \subseteq \mathbb{R}$. Prove using Darboux's theorem, or any other method, that $g'(x)$ has no jump discontinuities on A . (7)

- (b) Let f and g be functions continuous on $[a, b]$ and differentiable on (a, b) . By using the parametric curve $(f(x), g(x))$, or otherwise, give a graphical description of the *Generalised Mean Value Theorem*. (3)